

Sushant Chadha

College Station, TX 77840 | (609) 613 - 8466 | chadhasushant7@outlook.com | US Citizen

Education and Coursework

Major: BSAE – Bachelor of Science: Aerospace Engineering
Texas A&M University - GPA: 3.88

Expected Graduation in May '28
College Station, TX

Minors in Astrophysics and Computer Science

Skills

- **Special Skills:** Developing automation for fluid and mechanical workflows on Slurm-based HPC systems
 - **Additional Skills:** SolidWorks, ANSYS, STAR-CCM+, C++, Python, Java, JavaScript, Autodesk Inventor and AutoCAD, Unreal Engine 5
-

Experience

Bush Combat Development Complex Student Researcher I

Bryan, Texas

06/2025 - 08/2025

- Translated ROS2 messages obtained from multiple ground vehicles for use in Foxglove Studio for testing.
- Analyzed multi-object tracking experiment data and applied learning to program calibration, improving target tracking efficiency by 20%.
- Developed HTML + JS frontend to improve camera feed viewing & latency, allowing for 2 extra test runs per test day.

Texas A&M SAE AERO Design Team

College Station, Texas

Aerodynamics, Systems and Control Product Owner

04/2026 - Current

- Led team of 12 members through trade studies of alternate planforms and thin-wing concepts, developing a biplane and canard that would improve scoring performance by 20%.
- Developed Python-based driver for two-way Fluid Structure Interaction simulations, using Ansys Mechanical APDL and STAR-CCM+ to analyze aeroelasticity of experimental main wing designs.
- Created workflow to simulate propeller placement impacts on aerodynamic efficiency using STAR-CCM+, quantifying impacts on trim angle and previously unknown lift gains.

Aerodynamics, Systems and Control Member

05/2025 - 04/2026

- Led team migration from CPU-only CFD simulation to an automated GPU-accelerated simulation workflow using STAR-CCM+ and Python, reducing total runtime by 8x and resource consumption by 3x on Linux HPC systems by automating node selection.
- Performed automated airfoil analysis using AeroSandBox and Google Sheet macros to analyze 100s of airfoils and wing shapes to balance lift, drag, and trim angle.
- Performed static and dynamic powertrain testing using a low-speed wind tunnel to validate electrical and aerodynamic Python models.
- Updated flight test checklists to be in compliance with novel planform and ensured checklists were complete at all flight tests.

Code Ninjas Princeton

Princeton, NJ

Senior Sensei

08/2022 - 06/2024, 06/2026 - 08/2026

- Tutored 200 kids on basics of JavaScript, Unity, and Roblox Lua in combination with a self-driven curriculum
 - Led summer camps of 15+ kids teaching cybersecurity, game development, and AI basics to children aged 8-15.
-

Awards – Additional Involvement/Interests

- **Awards:** Eagle Scout, 5th/35 Teams – SAE Aero Design East, 5x Winner – Space Settlement Design Competitions
- **Additional Involvement/Interests:** Photography, Nature Preservation, Hiking